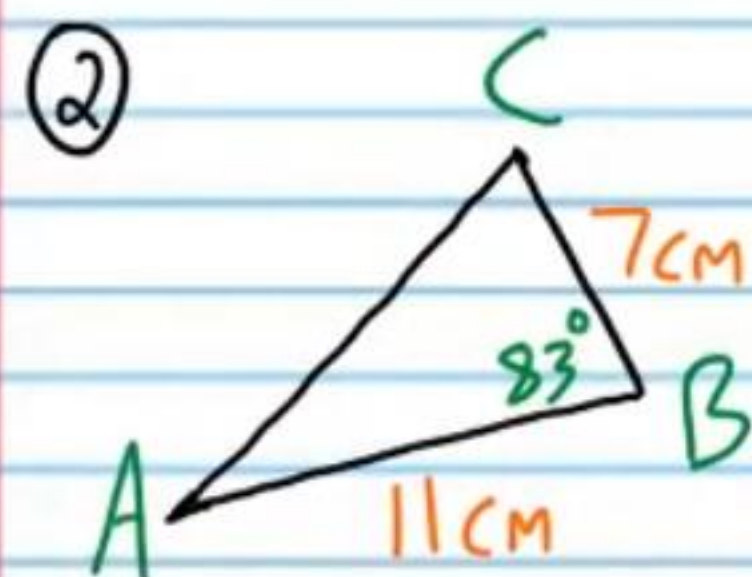
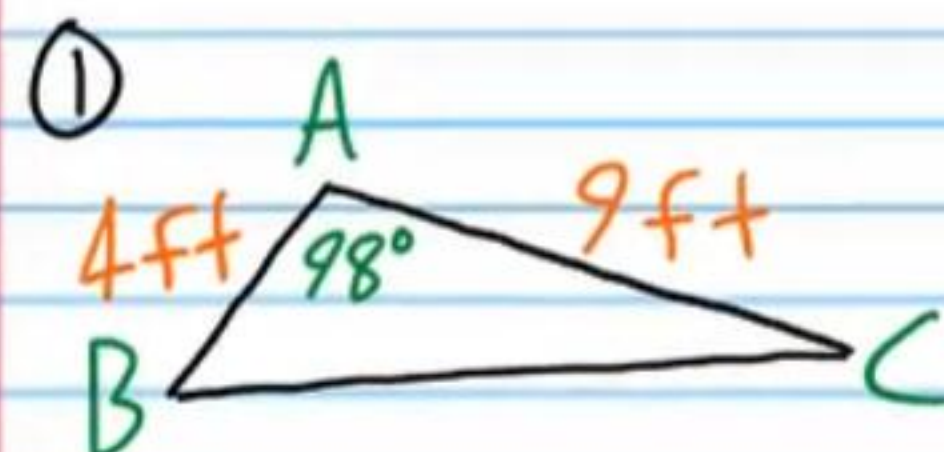
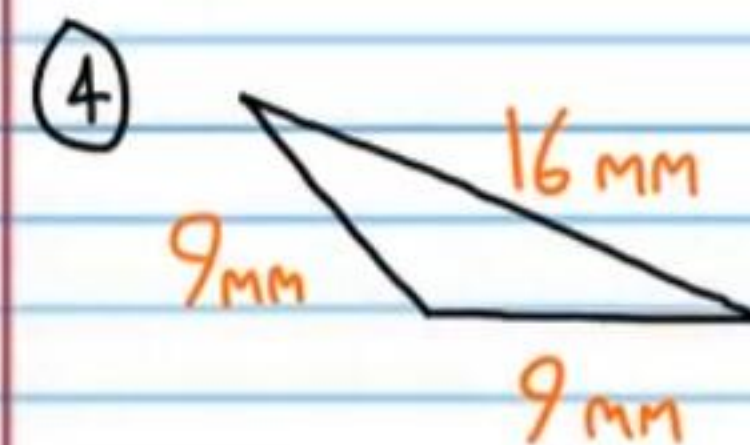


Applications of Trigonometry (Homework)

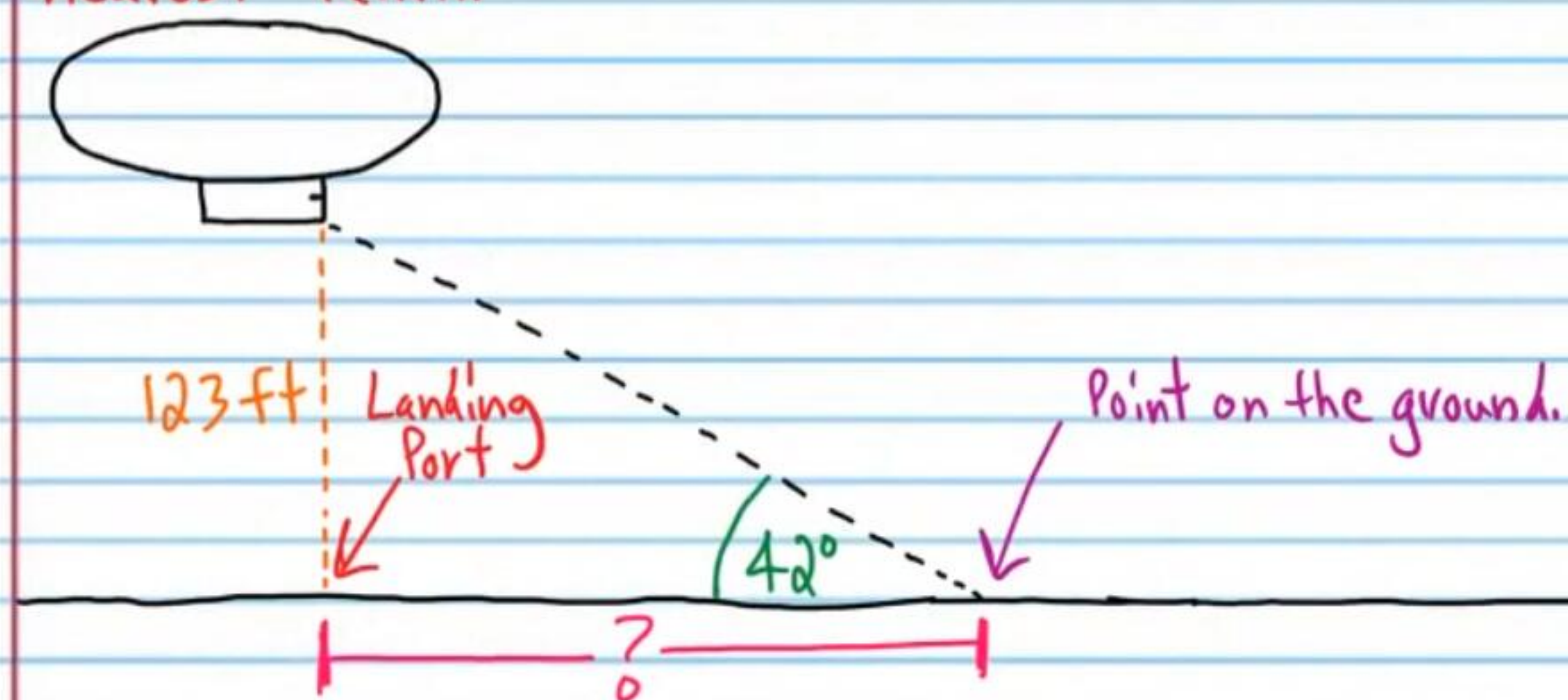
Find the area of the triangles below. Round to the nearest tenth.



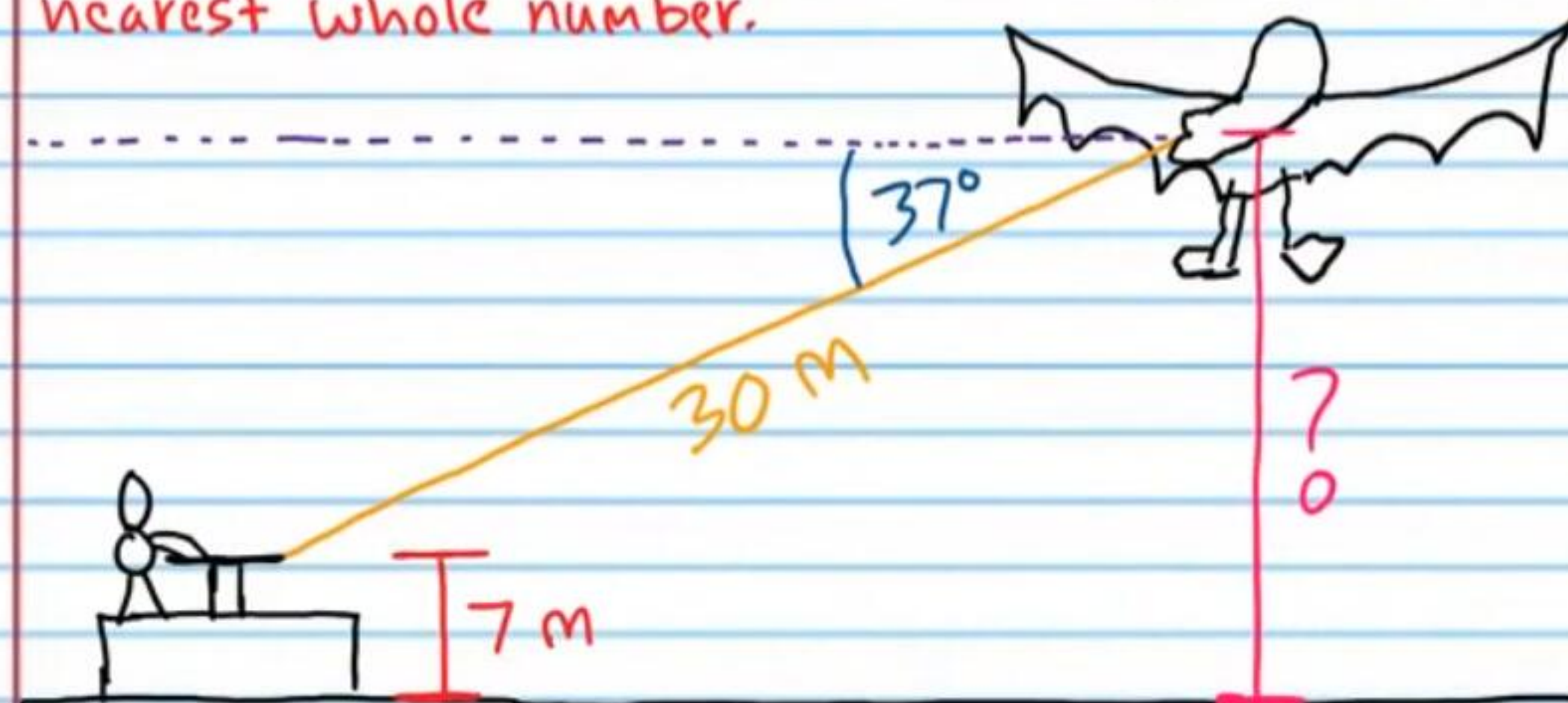
③ $a = 6\text{ in}$, $b = 8\text{ in}$, $C = 131^\circ$



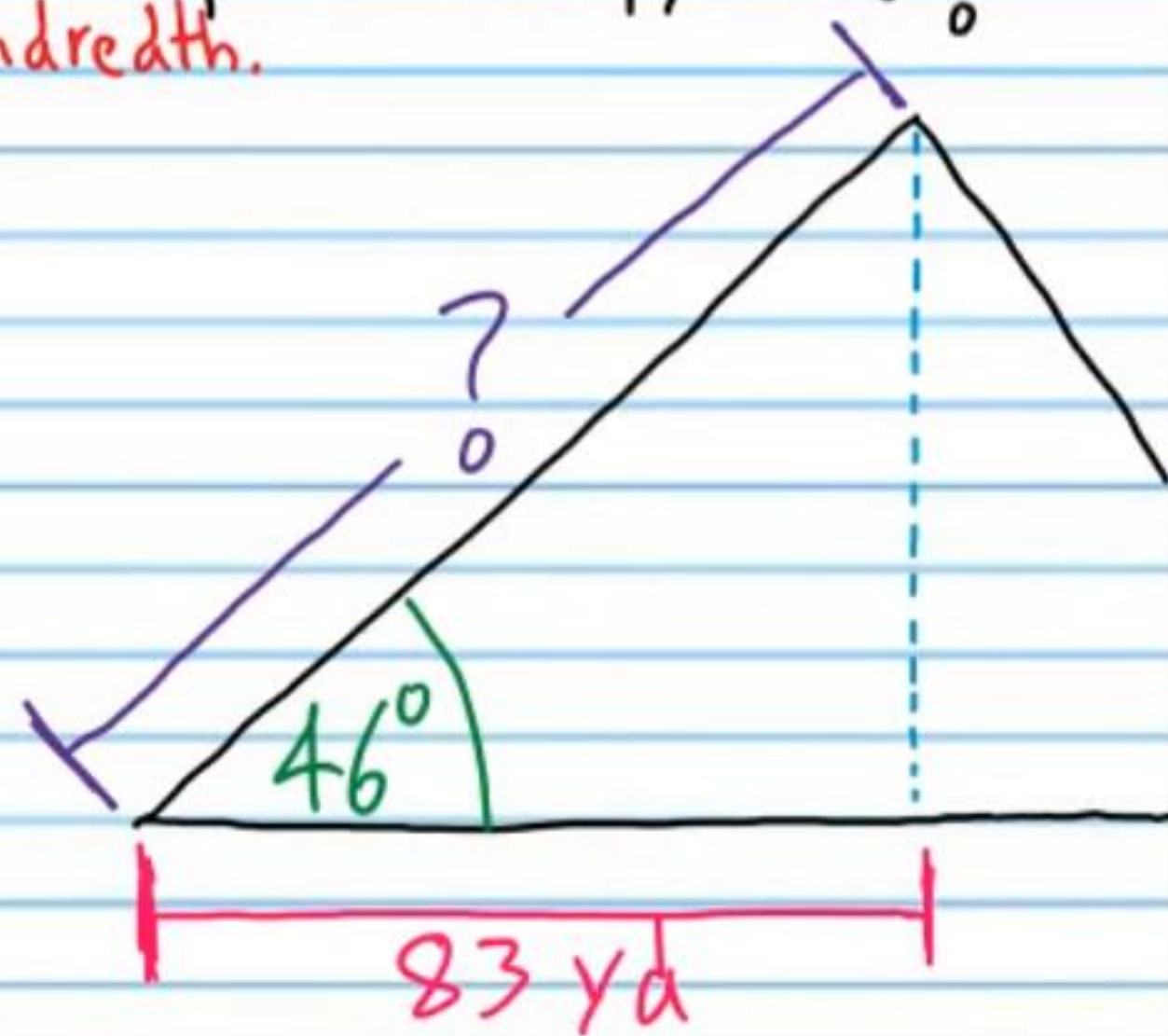
⑥ The angle of elevation from a point on the ground to a blimp is 42° . If the altitude of the blimp is 123 feet, what is the horizontal distance from the point on the ground to the landing port? Round to the nearest tenth.



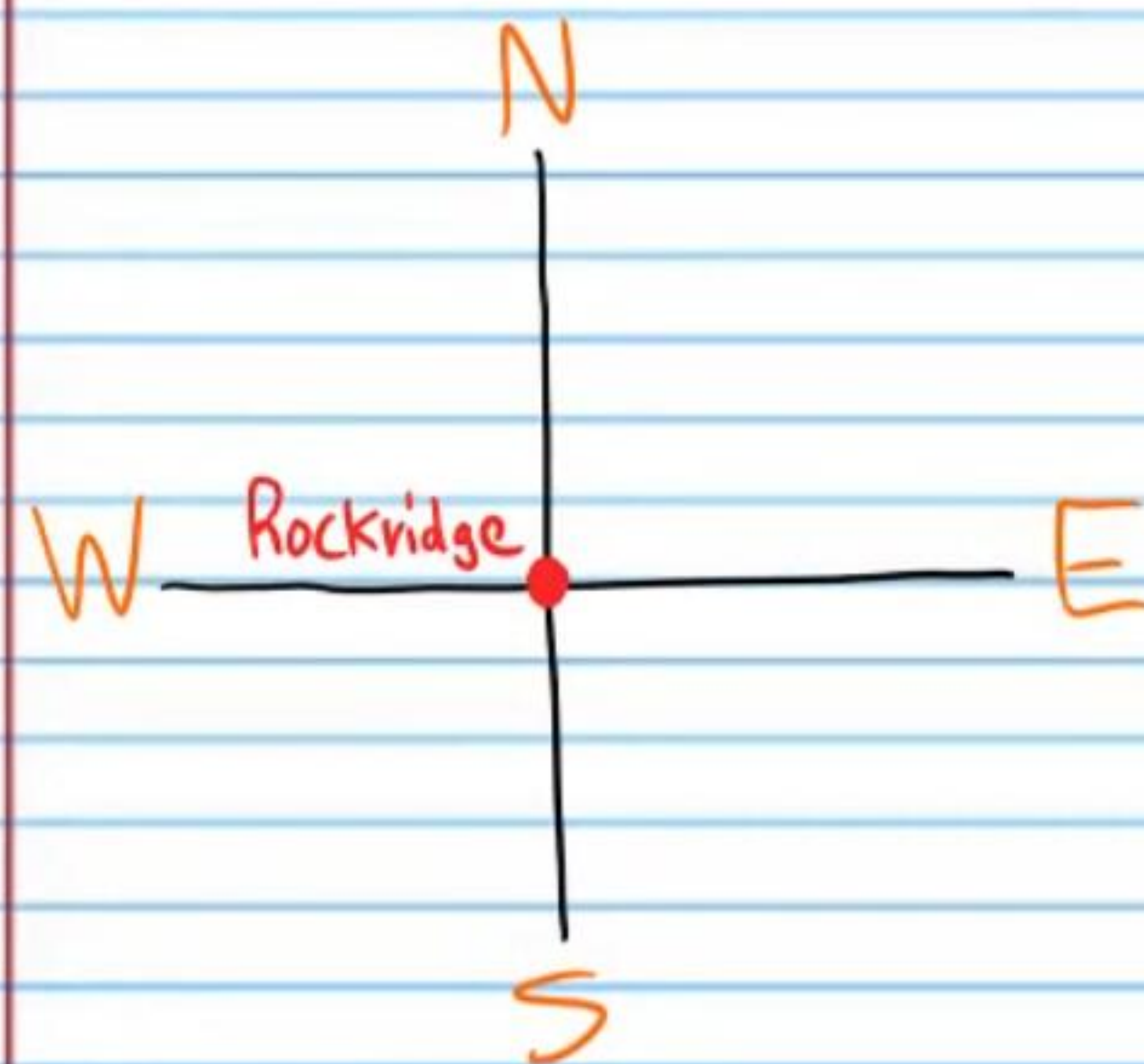
⑦ A knight shot a dragon with a harpoon tied to a rope. The harpoon gun is 7 meters off the ground and the rope is 30 meters long. What is the dragon's altitude if the rope makes a 37° angle of depression from the horizontal? Round to the nearest whole number.



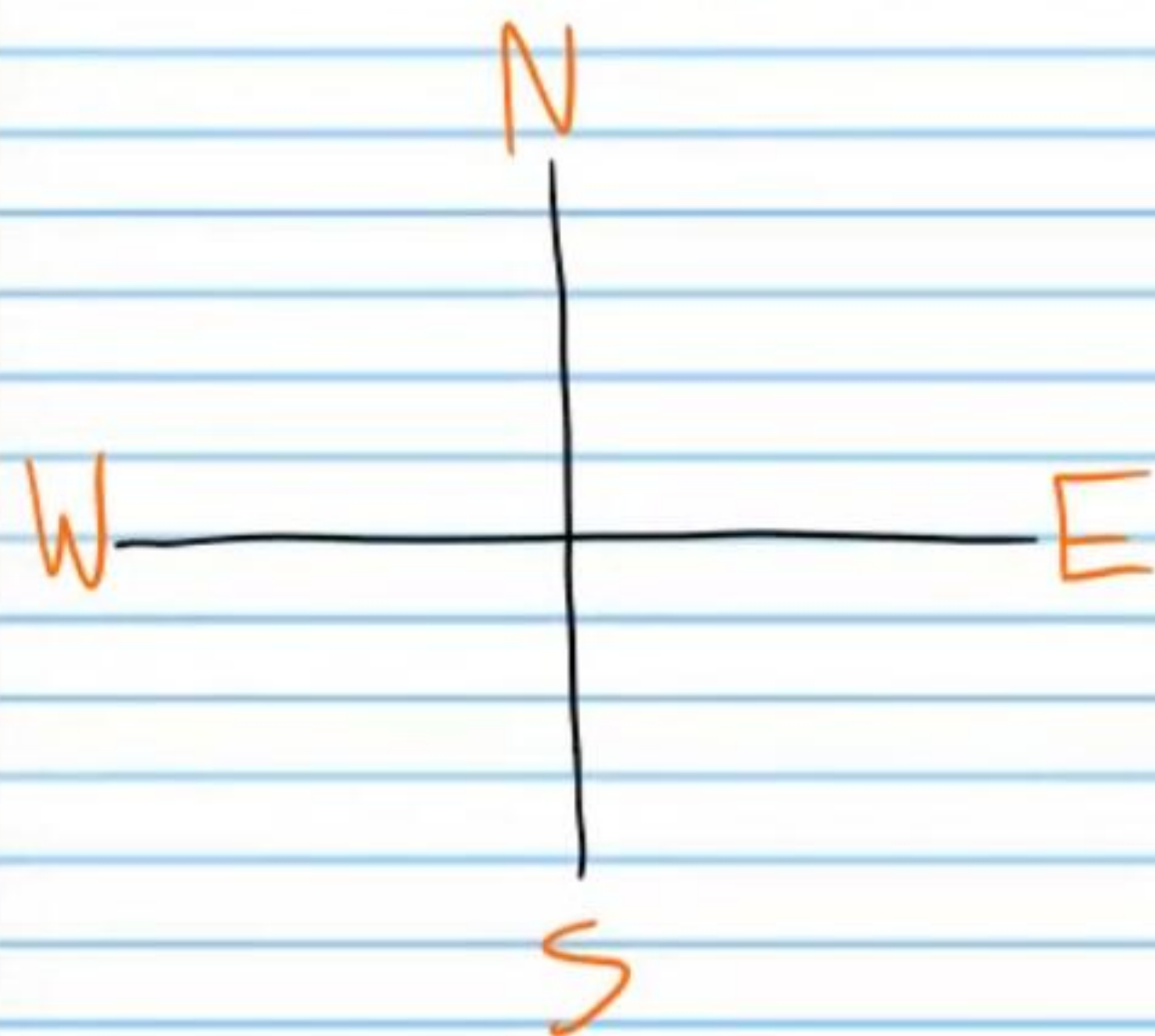
⑧ The angle of elevation from the side of a pyramid to the top of the pyramid is 46° . If the horizontal distance from the side of the pyramid to the center is 83 yards, how far would a person have to walk to get to the top of the pyramid?
Round to the nearest hundredth.



⑨ A map shows that a person in the city of Rockridge must hike 11 kilometers on a heading 29 degrees east of south ($S 29^\circ E$) to arrive at Melbourne Springs. If a person wants to hike from Rockridge to the town of Lawton, he must walk 6 kilometers on a heading 70 degrees east of south ($S 70^\circ E$). How far apart are Melbourne Springs and the town of Lawton?

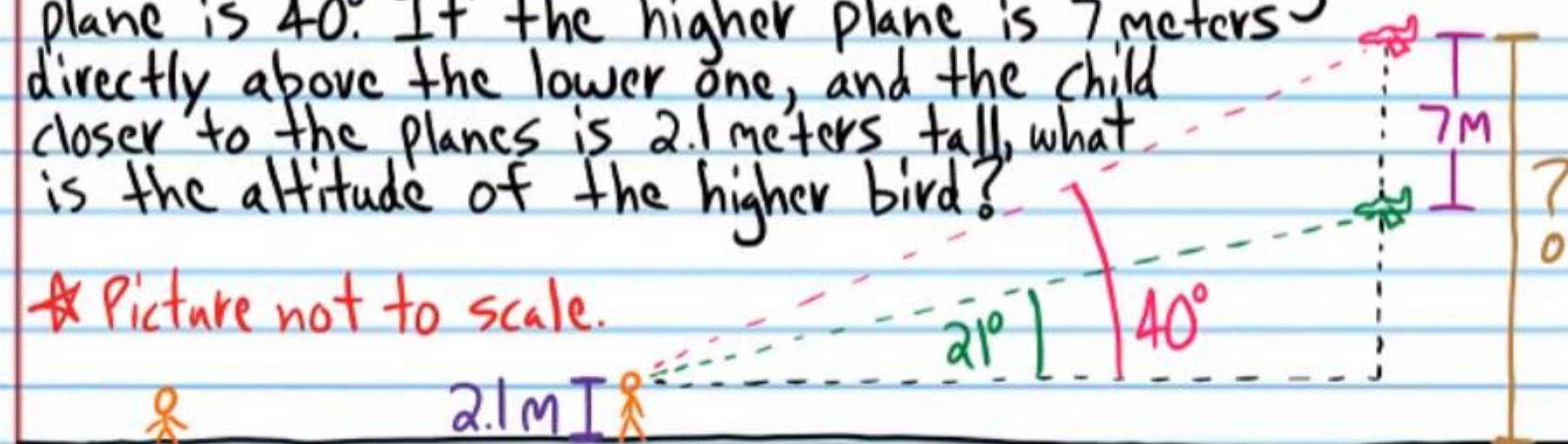


⑩ A gps device (global positioning system) shows that a bicyclist must ride 23 miles on a bearing 20 degrees north of west ($W20^{\circ}N$) to get to Arlington Beach. The device also shows that the bicyclist must ride 50 miles on a bearing 40 degrees south of west ($W40^{\circ}S$) to get to her friend Lydia's house. How far apart are Arlington Beach and Lydia's house?

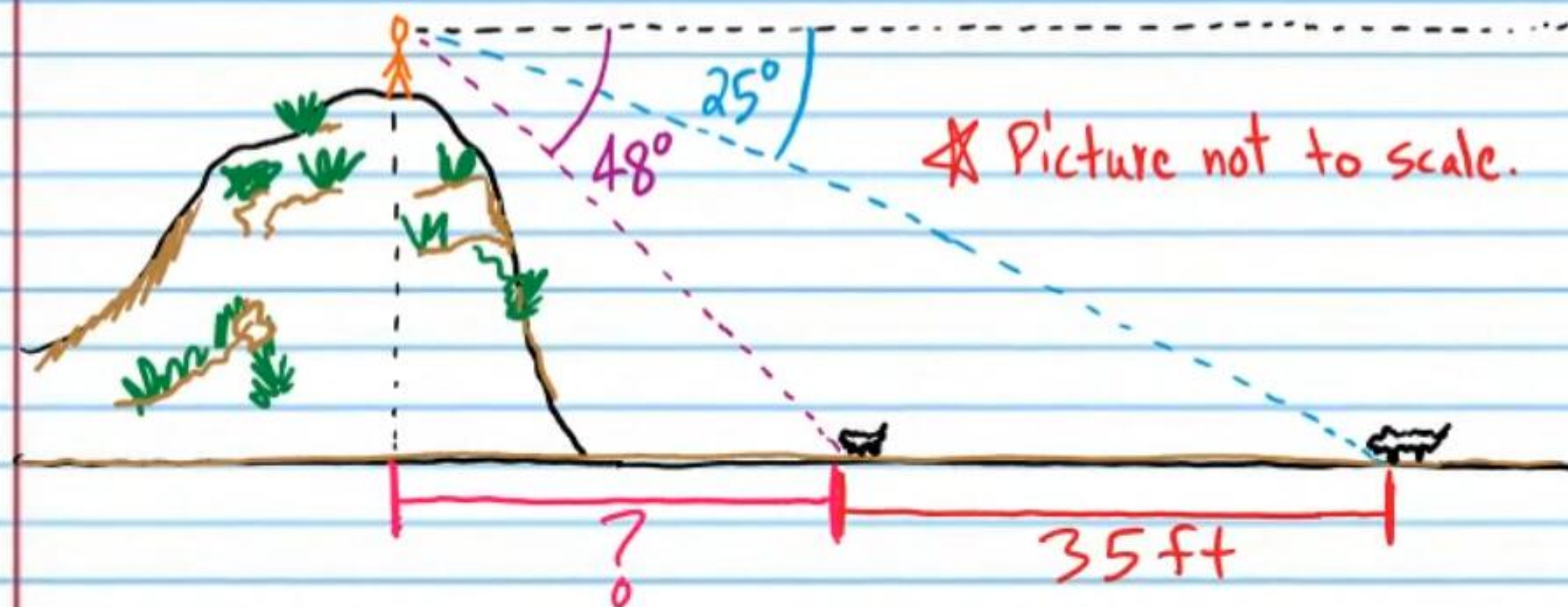


⑪ Two children are flying remote control planes. The angle of elevation from the child closer to the planes to the lower plane is 21° . The angle of elevation from the same child to the higher plane is 40° . If the higher plane is 7 meters directly above the lower one, and the child closer to the planes is 2.1 meters tall, what is the altitude of the higher bird?

* Picture not to scale.



⑫ A man standing on a hill looks down at his dog at ground level with an angle of depression of 48° and his other dog with an angle of depression of 25° . If the dogs are 35 feet apart, what is the horizontal distance from the person to the closer dog?



Homework Answers

① 17.8 ft^2

⑫ 25.33 ft

② 38.2 cm^2

③ 18.1 in^2

④ 33 mm^2

⑤ 38.9 km^2

⑥ 136.6 ft

⑦ 25 m

⑧ 119.48 yd

⑨ 7.57 km

⑩ 43.35 mi

⑪ 15 m